

Lab Sheet 5

Modeling Biological Spiking Neurons using the Izhikevich Model**Learning outcomes.**

By the end of this lab, you will be able to:

1. Implement the **Izhikevich spiking neuron model**
2. Simulate different **biological neuron firing patterns**
3. Analyze **membrane potential dynamics**
4. Compute **spike statistics**
5. Compare the Izhikevich neuron with the **Leaky Integrate-and-Fire model from Lab 4**

Background

However, real biological neurons exhibit much richer behaviour, including:

- tonic spiking
- bursting
- spike frequency adaptation
- phasic spiking
- rebound spikes

Izhikevich neuron model was introduced as a computationally efficient model capable of reproducing many biological firing patterns while remaining simple to simulate.

This model uses **two differential equations** and a **reset condition** to simulate neuron dynamics.

Izhikevich Neuron Model

The neuron is described by the following equations:

Membrane potential

$$\frac{dv}{dt} = 0.04v^2 + 5v + 140 - u + I$$

Recovery variable

$$\frac{du}{dt} = a(bv - u)$$

Spike Reset Rule

When the membrane potential reaches the spike threshold:

$$v \geq 30 \text{ mV}$$

Then:

$$\begin{aligned} v &\leftarrow c \\ u &\leftarrow u + d \end{aligned}$$

This reset represents the neuron firing a spike and returning to a resting state.

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4. Simulation Setup

The simulation uses the following parameters.

Parameter	Value
Total simulation time	1000 ms
Time step	0.1 ms
Input current	constant

5. Neuron Types

Different parameter values produce different neuron behaviours.

Regular Spiking Neuron

$a = 0.02$
 $b = 0.2$
 $c = -65$
 $d = 8$

Fast Spiking Neuron

$a = 0.1$
 $b = 0.2$
 $c = -65$
 $d = 2$

Bursting Neuron

$a = 0.02$
 $b = 0.2$
 $c = -50$
 $d = 2$

These neuron types correspond to different biological firing patterns observed in cortical neurons.

Lab Tasks

You will implement and analyze the Izhikevich neuron model.

Complete the missing code sections marked with **TODO comments**.

TODO 1:

```
# Compute dv and du using the Izhikevich equations
# dv = ...
# du = ...

# Update state variables using Euler method
# v_trace[i] = ...
# u_trace[i] = ...
```

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TODO 2:

```
# Implement spike reset:
# if v_trace[i] >= 30:
#   ...
#   ...
#   ...
```

TODO 3:

```
# Simulate the fast spiking neuron using FAST_SPIKING parameters
# Plot membrane potential
# Print spike count and firing rate
```

TODO 4:

```
# Simulate the bursting neuron using BURSTING parameters
# Plot membrane potential
# Print spike count and firing rate
```

TODO 5:

```
# Compute and print inter-spike intervals for the regular spiking neuron
```

TODO 6 – F-I Curve Analysis

```
# Use the Regular Spiking neuron parameters.
# Vary the input current from:
# 0 → 20
# For each current value:
# simulate the neuron
# compute the firing rate
# store the result
# Plot the F–I curve.
```

TODO 7 – Short Comparison (3–5 sentences)

```
# Compare the Izhikevich neuron with the LIF neuron from Lab 4.
# Discuss:
# - biological realism
# - number of firing patterns
# - computational cost
# - when each model is useful
```

Submission:

Rename the Lab 5 folder with your studentID then create a zip file and upload.